

2.2.2 Derivation of the Verlet Method

Now let us consider the discrete version of “minimizing the action”. We work on the time interval $[0, \tau]$. Consider the $\nu + 1$ points $\mathbf{q}_0$ to $\mathbf{q}_\nu$ in configuration space

$$\hat{\Gamma} = (\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_\nu).$$

We must first define the action for such a discrete path. One perspective is that we first formulate a discrete version of the Lagrangian in terms of only the point set $\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_\nu$, but this somehow requires that we define velocities. Noting that

$$\dot{\mathbf{q}}(t) \approx \frac{\mathbf{q}(t+h) - \mathbf{q}(t)}{h},$$

we are led to consider the approximation at time level n

$$\mathbf{v}_n \stackrel{\text{def}}{=} \frac{\mathbf{q}_{n+1} - \mathbf{q}_n}{h},$$

and thus, by Riemann summation

$$\mathcal{A}_L \approx \hat{\mathcal{A}} \stackrel{\text{def}}{=} \sum_{n=0}^{\nu-1} L\left(\mathbf{q}_n, \frac{\mathbf{q}_{n+1} - \mathbf{q}_n}{h}\right) h.$$

At first glance, it looks like this is a crude approximation, since we have employed a one-sided difference, however, let us proceed anyway to see the implications of this choice. In the case of a mechanical system with Lagrangian $L(\mathbf{q}, \mathbf{v}) = \dot{\mathbf{v}}^T \mathbf{M} \mathbf{v} / 2 + U(\mathbf{q})$, we then have

$$\hat{\mathcal{A}} = \sum_{n=0}^{\nu-1} \left[\frac{(\mathbf{q}_{n+1} - \mathbf{q}_n)^T \mathbf{M} (\mathbf{q}_{n+1} - \mathbf{q}_n)}{2h^2} - U(\mathbf{q}_n) \right] h.$$

We should think of $\hat{\mathcal{A}}$ as a function of all the positions $\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_\nu$ defining the discrete path. Critical points of this function satisfy

$$\nabla \hat{\mathcal{A}} = 0,$$

where the gradient must be taken with respect to all configurational points on the path (and all coordinates). This condition leads to the equations

$$\frac{\partial \hat{\mathcal{A}}}{\partial \mathbf{q}_n} = 0, \quad n = 1, \dots, \nu,$$